

Chapter 3. Sums over Poisson processes

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□ Remarks

Def $(\Omega_1, \mathcal{F}_1)$ $(\Omega_2, \mathcal{F}_2)$: measurable space

$f: \Omega_1 \rightarrow \Omega_2$ where $f^{-1}(E) \in \mathcal{F}_1$ for $\forall E \in \mathcal{F}_2$

Then f is a measurable function

Let μ be a measure on $(\Omega, \mathcal{F})$ we will define $\int f d\mu$ where f is measurable function

- | | | |
|-------------------------|---|--|
| ① Simple functions | } | with this 4 steps
we can define $\int f d\mu$. |
| ② Bounded functions | | |
| ③ Nonnegative functions | | |
| ④ General functions | | |

Step 1.

ϕ : simple function if $\phi(\omega) = \sum_{i=1}^n a_i 1_{A_i}$ (A_i : disjoint, $\mu(A_i) < \infty$)

If ϕ : simple function $\rightarrow \int \phi d\mu = \sum_{i=1}^n a_i \mu(A_i)$

Step 2,

$\mu(E) < \infty$ and f is bounded function that vanishes on E^c .

If ϕ, ψ are simple functions s.t., $\phi \leq f \leq \psi$

$$\int \phi d\mu \leq \int f d\mu \leq \int \psi d\mu$$

So, define $\int f d\mu = \sup_{\phi \leq f} \int \phi d\mu = \inf_{\psi \geq f} \int \psi d\mu$

Step 3.

If $f \geq 0$

$$\int f d\mu = \sup \left\{ \int h d\mu \mid 0 \leq h \leq f, h: \text{bounded} \ \& \ \mu(\{x \mid h(x) > 0\}) < \infty \right\}$$

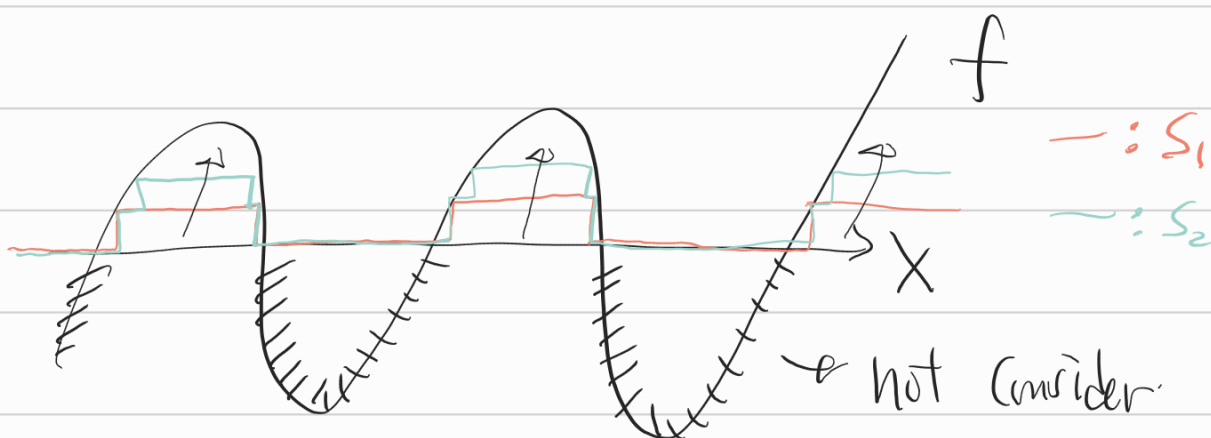
Step 4.

Let $f^+(x) = \max(f(x), 0)$, $f^-(x) = -\min(f(x), 0)$

$\rightarrow f(x) = f^+(x) - f^-(x)$ & $f^+, f^- \geq 0$

$$\int f d\mu = \int f^+ d\mu - \int f^- d\mu$$

(We say f is integrable if $\int |f| d\mu < \infty$)



<Constructing the simple functions>

I Means, variances, and distributions

Through this chapter we will devote to sums of the form

$$\Sigma := \sum_{X \in \Pi} f(X) \quad (f: S \text{ (state space of } \Pi) \rightarrow \mathbb{R})$$

Ex) Suppose that stars in some region of space are scattered according to a Poisson process Π & stars are of the same mass

$$\Rightarrow \text{Gravitational potential at a point } x: \Sigma = \sum_{X \in \Pi} |x - X|^{-1}$$

To find the distribution of Σ , we calculate it for simple functions f and then to extend the result to more complicated f .

Let $f: S \rightarrow \mathbb{R}$ which takes only a finite number of non-zero values $f_1, f_2, \dots, f_k$ and is zero outside & mean measure μ of Π is finite.

$A_j = \{X \mid f(X) = f_j\}$: measurable & $m_j = \mu(A_j) < \infty$ & A_j are disjoint
 $\therefore N_j = N(A_j)$ are independent, $N_j \sim P(m_j)$

$$\Sigma = \sum_{X \in \Pi} f(X) = \sum_{j=1}^k f_j N_j$$

To determine the distribution of Σ , characteristic function is a good option!

For any real or complex θ ,

$$\begin{aligned}
 E(e^{\theta \Sigma}) &= \prod_{j=1}^K E(e^{\theta f_j N_j}) \\
 &= \prod_{j=1}^K \sum_{\tilde{n}=0}^{\infty} e^{\theta f_j \tilde{n}} \cdot \frac{m_j^{\tilde{n}} \cdot e^{-m_j}}{\tilde{n}!} \quad (\because N_j \sim P(m_j)) \\
 &= \prod_{j=1}^K \sum_{\tilde{n}=0}^{\infty} \frac{(e^{\theta f_j m_j})^{\tilde{n}} \cdot e^{-e^{\theta f_j m_j}}}{\tilde{n}!} \cdot e^{(e^{\theta f_j} - 1) m_j} \\
 &= \prod_{j=1}^K \exp\{(e^{\theta f_j} - 1) m_j\} \\
 &= \exp\left\{\sum_{j=1}^K \int_{A_j} (e^{\theta f(x)} - 1) \mu(dx)\right\} \\
 &= \exp\left\{\int_S (e^{\theta f(x)} - 1) \mu(dx)\right\} \quad (\because f \equiv 0 \text{ outside the union of the } A_j)
 \end{aligned}$$

(Case 1) f can take positive and negative values

$\Rightarrow$ Take θ pure imaginary number

$$\therefore E(e^{it\Sigma}) = \exp\left\{\int_S (e^{itf(x)} - 1) \mu(dx)\right\} \quad (t: \text{real})$$

(Will prove in next section) (Campbell's thm)

For any measurable f , $\Sigma := \sum_{x \in \Pi} f(x)$ absolutely convergent a.s

$\Leftrightarrow \exp\left\{\int_S (e^{itf(x)} - 1) \mu(dx)\right\}$ converges absolutely ($t \neq 0$)

Then, $E(e^{it\Sigma}) = \exp\left\{\int_S (e^{itf(x)} - 1) \mu(dx)\right\}$ ($t: \text{real}$) holds.

(case 2) f takes only positive values
 $\Rightarrow$ Take θ real and negative

$$\therefore E(e^{-u\Sigma}) = \exp \left\{ - \int_S (1 - e^{-uf(x)}) \mu(dx) \right\} \quad (\#)$$

If the integral diverges, $\Sigma = \infty$ with probability 1

(#) takes trivial form $0=0$

$\therefore$ This form is true without restriction

Also we can expand $E(e^{\theta\Sigma}) = \exp \left\{ \int_S (e^{\theta f(x)} - 1) \mu(dx) \right\}$ as a power series

$\Rightarrow$ Comparing coefficients induces this results

$$\left\{ \begin{array}{l} E(\Sigma) = \int_S f(x) \mu(dx) \\ \text{Var}(\Sigma) = \int_S f(x)^2 \mu(dx) \end{array} \right.$$

$$\begin{aligned} \therefore E(e^{\theta\Sigma}) &= E\left(1 + \theta\Sigma + \frac{1}{2}\theta^2\Sigma^2 + \dots\right) \\ &= 1 + E(\Sigma)\theta + \frac{1}{2}E(\Sigma^2)\theta^2 + \dots \end{aligned}$$

$$\begin{aligned} \exp \left\{ \int_S (e^{\theta f(x)} - 1) \mu(dx) \right\} &= \exp \left\{ \int_S \left(\theta f(x) + \frac{1}{2}\theta^2 f(x)^2 + \dots \right) \mu(dx) \right\} \\ &= e^{\int_S f(x) \mu(dx) \theta} \cdot e^{\frac{1}{2} \int_S f(x)^2 \mu(dx) \theta^2} \cdot \dots \\ &= \left(1 + \int_S f(x) \mu(dx) \theta + \frac{1}{2} \left(\int_S f(x) \mu(dx) \right)^2 \theta^2 + \dots \right) \\ &\quad \cdot \left(1 + \frac{1}{2} \int_S f(x)^2 \mu(dx) \theta^2 + \dots \right) \cdot \dots \end{aligned}$$

$$\theta \text{ coefficient: } E(\Sigma) = \int_S f(x) \mu(dx)$$

$$\left. \begin{aligned} \theta^2 \text{ coefficient: } \frac{1}{2} E(\Sigma^2) &= \frac{1}{2} \left(\int_S f(x) \mu(dx) \right)^2 + \frac{1}{2} \int_S f(x)^2 \mu(dx) \\ \therefore \text{var}(\Sigma) &= E(\Sigma^2) - E(\Sigma)^2 = \int_S f(x)^2 \mu(dx) \quad (1/1) \end{aligned} \right\}$$

If f_1, f_2 are real-valued functions, $\theta f \rightarrow \theta_1 f_1 + \theta_2 f_2$

$$E(e^{\theta_1 \Sigma_1 + \theta_2 \Sigma_2}) = \exp \left\{ \int_S (e^{\theta_1 f_1(x) + \theta_2 f_2(x)} - 1) \mu(dx) \right\}$$

It determines joint distribution of $\Sigma_1 = \sum_{x \in \pi} f_1(x)$, $\Sigma_2 = \sum_{x \in \pi} f_2(x)$

$$E(\Sigma_1 \Sigma_2) = \int f_1 d\mu \int f_2 d\mu + \int f_1 f_2 d\mu$$

$\therefore$ compare coefficient of $\theta_1 \theta_2$

$$\theta_1 \theta_2 \text{ coefficient of } E(e^{\theta_1 \Sigma_1 + \theta_2 \Sigma_2}) = \Sigma_1 \Sigma_2$$

$$\begin{aligned} \exp \left\{ \int_S (e^{\theta_1 f_1(x) + \theta_2 f_2(x)} - 1) \mu(dx) \right\} &= 1 + \int_S (e^{\theta_1 f_1(x) + \theta_2 f_2(x)} - 1) \mu(dx) + \frac{1}{2!} \left(\int_S (e^{\theta_1 f_1(x) + \theta_2 f_2(x)} - 1) \mu(dx) \right)^2 + \dots \\ &= 1 + \int_S (\theta_1 f_1(x) + \theta_2 f_2(x) + \theta_1 \theta_2 f_1 f_2 + \dots) \mu(dx) + \frac{1}{2} \left(\int_S (\theta_1 f_1(x) + \theta_2 f_2(x) + \theta_1 \theta_2 f_1 f_2 + \dots) \mu(dx) \right)^2 + \dots \end{aligned}$$

$$\Rightarrow \theta_1 \theta_2 \text{ coefficient of } \exp \left\{ \int_S (e^{\theta_1 f_1(x) + \theta_2 f_2(x)} - 1) \mu(dx) \right\} = \int_S f_1 f_2 d\mu + \int_S f_1 d\mu \int_S f_2 d\mu$$

$$\therefore \text{covariance of } \Sigma_1 \text{ and } \Sigma_2 : E(\Sigma_1 \Sigma_2) - E(\Sigma_1) E(\Sigma_2) = \int f_1 f_2 d\mu$$

We can also write $E(\Sigma_1 \Sigma_2)$ as

$$E\left(\sum_{x_1, x_2 \in \pi} f_1(x_1) f_2(x_2)\right) = E\left(\underbrace{\sum_{x \in \pi} f_1(x) f_2(x)}_{= \int f_1 f_2 d\mu}\right) + E\left(\sum_{x_1 \neq x_2} f_1(x_1) f_2(x_2)\right)$$

$$\therefore E\left(\sum_{x_1 \neq x_2} f_1(x_1) f_2(x_2)\right) = E\left(\sum_{x_1 \in \pi} f_1(x_1)\right) E\left(\sum_{x_2 \in \pi} f_2(x_2)\right)$$

Also this identity generalizes to $n(z)$ functions

$$E\left(\sum_{\substack{x_1, \dots, x_n \in \Pi \\ x_j \text{ distinct}}} f_1(x_1) f_2(x_2) \cdots f_n(x_n)\right) = \prod_{j=1}^n E\left(\sum_{x_j \in \Pi} f_j(x_j)\right)$$

[2] Campbell's Theorem

<Campbell's Thm>

Let Π be a Poisson process on S with mean measure μ , $f: S \rightarrow \mathbb{R}$ measurable

Then the sum $\Sigma = \sum_{x \in \Pi} f(x)$ is absolutely convergent a.s. $\Leftrightarrow \int_S \min(|f(x)|, 1) \mu(dx) < \infty$

If this condition holds, then $E(e^{\theta \Sigma}) = \exp\left\{\int_S (e^{\theta f(x)} - 1) \mu(dx)\right\}$ for any complex θ for which the integral on the right converges, and in particular whenever θ is pure imaginary.

Moreover $E(\Sigma) = \int_S f(x) \mu(dx)$ in the sense that $E(\Sigma)$ exists $\Leftrightarrow$ integral converges and they are then equal. If $\int_S f(x) \mu(dx)$ converges, then $\text{Var}(\Sigma) = \int_S f(x)^2 \mu(dx)$ finite or infinite.

pf)

Any positive measurable function f may be expressed as the limit of an increasing sequence (f_j) of simple functions.

Take $\theta = -u(<0)$, $\Sigma_j = \sum f_j(x)$

$$E(e^{-u \Sigma}) = \lim_{j \rightarrow \infty} E(e^{-u \Sigma_j}) \quad (\because \text{by M.C.T})$$

$$= \lim_{j \rightarrow \infty} \exp\left\{-\int (1 - e^{-u f_j(x)}) \mu(dx)\right\} \quad (\because f_j \text{ is simple})$$

$$= \exp\left\{-\int (1 - e^{-u f(x)}) \mu(dx)\right\} \quad (\because \text{by M.C.T})$$

If $\int_S \min(|f(x)|, 1) \mu(dx) < \infty$ & $u > 0$

Let $A_1 = \{x \in S \mid |f(x)| < 1\}$, $A_2 = \{x \in S \mid |f(x)| \geq 1\}$

$$\begin{aligned} \int_S (1 - e^{-u|f(x)|}) \mu(dx) &= \int_{A_1} (1 - e^{-u|f(x)|}) \mu(dx) + \int_{A_2} (1 - e^{-u|f(x)|}) \mu(dx) \\ &\leq \int_{A_1} (1 - e^{-u|f(x)|}) \mu(dx) + \int_{A_2} 1 \mu(dx) \\ &\leq \int_{A_1} M \cdot u|f(x)| \mu(dx) + \int_{A_2} 1 \mu(dx) \quad (\because 1 - e^{-x} \leq x) \\ &< \infty \end{aligned}$$

& $u \rightarrow 0 \Rightarrow \int_S (1 - e^{-u|f(x)|}) \mu(dx) \rightarrow 0$

$\therefore \Sigma$: finite random variable ($P(\Sigma < \infty) = 1$)

Then $E(e^{\theta \Sigma})$, $\exp\{\int_S (e^{\theta f(x)} - 1) \mu(dx)\}$ analytic of θ in $\operatorname{Re} \theta \leq 0$

$\Rightarrow$ By analytic continuation $E(e^{\theta \Sigma}) = \exp\{\int_S (e^{\theta f(x)} - 1) \mu(dx)\}$ for $\operatorname{Re} \theta \leq 0$

If $\int_S \min(|f(x)|, 1) \mu(dx) = \infty \rightarrow \int_S (1 - e^{-u|f(x)|}) \mu(dx) = \infty$ for $\forall u > 0$

$\therefore E(e^{-u \Sigma}) = 0 \rightarrow P(\Sigma = \infty) = 1$

We can also do $E(\Sigma) = \int_S f(x) \mu(dx)$, $\operatorname{Var}(\Sigma) = \int_S f(x)^2 \mu(dx)$ exactly.

Similar arguments.

$\therefore$ So we proved Campbell's Theorem for $f \geq 0$

To complete the proof for general function f we define f^+, f^-
 $f^+ := \max(f, 0)$, $f^- := \max(-f, 0)$

Σ absolutely converges $\Leftrightarrow \Sigma_+ = \sum_{x \in \Pi} f^+(x)$, $\Sigma_- = \sum_{x \in \Pi} f^-(x)$ converges

Let Π_+, Π_- : restrictions of Π to $S_+ = \{f \geq 0\}$, $S_- = \{f < 0\}$

$$\Sigma_+ = \sum_{x \in \Pi} f^+(x) = \sum_{x \in \Pi_+} f(x), \quad \Sigma_- = \sum_{x \in \Pi} f^-(x) = - \sum_{x \in \Pi_-} f(x)$$

$$\therefore \Sigma = \sum_{x \in \Pi} f(x) \text{ absolutely convergent a.s.} \Leftrightarrow \int_S \min(|f(x)|, 1) \mu(dx) < \infty$$

Let θ : pure imaginary number

$$\begin{aligned} E(e^{\theta \Sigma}) &= E(e^{\theta \Sigma_+ - \theta \Sigma_-}) \\ &= E(e^{\theta \Sigma_+}) E(e^{-\theta \Sigma_-}) \quad (\text{restriction to disjoint sets}) \\ &\quad (\because \Pi_+, \Pi_- \text{ independent Poisson process}) \\ &= \exp \left\{ \int (e^{\theta f^+} - 1) d\mu \right\} \exp \left\{ \int (e^{-\theta f^-} - 1) d\mu \right\} \\ &= \exp \left\{ \int (e^{\theta f^+} + e^{\theta f^-} - 2) d\mu \right\} \\ &= \exp \left\{ \int (e^{\theta f} - 1) d\mu \right\} \quad (\because f \geq 0 \Leftrightarrow f^- = 0) \end{aligned}$$

$\therefore E(e^{\theta \Sigma}) = \exp \left\{ \int (e^{\theta f} - 1) d\mu \right\}$ at least for imaginary θ and its extension to all complex θ for which the integral converges, ($\because$ Analytic continuation) □

Cor 3.1 If $f_1, \dots, f_n$ satisfy $\int_S \min(|f_i(x)|, 1) \mu(dx) < \infty$, so that $\Sigma_j = \sum_{x \in \Pi} f_j(x)$ converges a.s. & $E(e^{it_1 \Sigma_1 + \dots + it_n \Sigma_n}) = \exp \left\{ \int_S (e^{it_1 f_1 + \dots + it_n f_n} - 1) \mu(dx) \right\}$

If f_j satisfy $\int_S f_j(x)^2 \mu(dx) < \infty$, then $\text{cov}(\Sigma_j, \Sigma_k) = \int_S f_j(x) f_k(x) \mu(dx)$

pf) $f \rightarrow t_1 f_1 + \dots + t_n f_n, \theta \rightarrow i$ in Campbell's Theorem □

[3] The characteristic functional

Def Characteristic functional (Laplace functional) of a random process
 $:= E(e^{-\sum f})$

$\Rightarrow E(e^{-\sum f}) = \exp\left\{-\int_S (1 - e^{-f(x)}) \mu(dx)\right\}$ (*) characterises the Poisson process
(special case of Campbell's thm setting $\theta = -1$)

Lemma $E[e^{uX+vY}] = E[e^{uX}]E[e^{vY}] \Leftrightarrow X, Y$ independent

pf) ($\Leftarrow$) is trivial so we will only prove ($\Rightarrow$).

Let $f(X, Y)$: Joint distribution of X, Y

Then we can get marginal distribution $f_X(x), f_Y(y)$ and we can

construct each random variable X', Y' has distribution $f_X(x), f_Y(y)$.

And when we define Joint distribution of X', Y' to $f_X(x), f_Y(y)$

$\Rightarrow X' \overset{d}{\sim} X, Y' \overset{d}{\sim} Y$ (equal in distribution) & X', Y' independent

$$E[e^{uX'+vY'}] = E[e^{uX'}]E[e^{vY'}] = E[e^{uX}]E[e^{vY}] = E[e^{uX+vY}]$$

By Moment generating function $M_X(t) := E[e^{tX}]$ uniqueness theorem

$$(X, Y) \overset{d}{\sim} (X', Y')$$

$\therefore X, Y$ independent. $\square$

Suppose that Π is any random countable subset of S , and define Σ_f for any function $f: S \rightarrow [0, \infty)$. Suppose that for some measure μ (*) holds for a certain class of functions $\mathcal{F}$

If $\mathcal{F}$ is rich enough to contain all functions that take on only a finite number of different values $f_1, f_2, \dots, f_k$

$$\Rightarrow E(e^{-\Sigma_f}) = \exp\left\{-\sum_{j=1}^k (1 - e^{-f_j}) m_j\right\} \quad \& \quad \Sigma_f = \sum_{j=1}^k f_j N(A_j)$$

$$(m_j = \mu(A_j), \quad A_j := \{x \mid f(x) = f_j\})$$

$$\therefore E(e^{-f_1 N(A_1)} \cdot e^{-f_2 N(A_2)} \cdots e^{-f_k N(A_k)}) = \prod_{j=1}^k E(e^{-f_j N(A_j)})$$

By lemma $N(A_j)$ are independent with Poisson distribution $P(m_j)$

$\therefore \Pi$ is a Poisson process with mean measure μ . ($\because$ definition)

★ To prove that a random countable set is a Poisson process, it therefore suffices to establish (*) for a function which takes on only a finite number of different values.

<Characterization of the homogeneous Poisson process in 1 dimension>

$\Rightarrow$ Random subset A of $\mathbb{R}$ which is a finite union of intervals, $N(A) \sim P(\lambda|A|)$ then the random set is a Poisson process with constant rate λ (was proved by Rényi in 1967)

It's surprising because it makes no assumption of independence of the $N(A_j)$ for disjoint A_j .

4 Rényi's Theorem

We want to generalize Rényi's original result.

① mean measure need not be a multiple of Lebesgue measure

② state space $\mathbb{R} \rightarrow \mathbb{R}^d$

③ avoidance function: $\alpha(A) = \mathbb{P}\{N(A)=0\} = \mathbb{P}\{\pi \cap A = \emptyset\}$

(ex) Poisson process with mean measure μ : $\alpha(A) = e^{-\mu(A)}$

↳ Opposite doesn't always have to be true!

④ rectangles in $\mathbb{R}^d$: $a, b \in \mathbb{R}^d$

$[a, b] = \{x = (x_1, \dots, x_d) \mid a_i \leq x_i \leq b_i \text{ } (i=1, \dots, d)\}$

<Rényi's Thm.>

Let μ : non-atomic measure on $\mathbb{R}^d$ & finite on bounded sets.

Let π : random countable subset of $\mathbb{R}^d$ s.t. whenever A is a finite union of rectangles $\mathbb{P}\{\pi \cap A = \emptyset\} = e^{-\mu(A)}$

Then π is a Poisson process with mean measure μ .

pf) Let k -cube $[a, b] : a_i = v_i 2^{-k}, b_i = (v_i + 1) 2^{-k} \ (1 \leq i \leq d, v_i : \text{integers})$

Let $C_1, C_2, \dots, C_n : k$ -cubes which are disjoint for fixed k

& $E(C) := \{ \pi \cap C = \emptyset \}$ (event that there are no points of the random set π in k -cube C)

$$\begin{aligned} \mathbb{P} \left\{ \bigcap_{r=1}^n E(C_r) \right\} &= \mathbb{P} \left\{ \pi \cap C_r = \emptyset \ (r=1, \dots, n) \right\} \\ &= \mathbb{P} \left\{ \pi \cap \left(\bigcup C_r \right) = \emptyset \right\} \\ &= \alpha \left(\bigcup C_r \right) \\ &= \prod_{r=1}^n e^{-\mu(C_r)} \quad (\because \text{assumption}) \end{aligned}$$

$\therefore$ Events $E(C)$ for k -cubes are independent with $\mathbb{P}\{E(C)\} = e^{-\mu(C)}$

For any bounded open set G , we want to compute the distribution of

$$N(G) = \# \{ \pi \cap G \}$$

$$\Rightarrow N(G) = \lim_{k \rightarrow \infty} N_k(G) \quad (N_k(G) : \# \text{ of } k\text{-cubes } C(CG) \text{ for which } E(C) \text{ does not occur})$$

($\because k \neq \emptyset \Rightarrow$ two distinct point of G lie in different k -cubes)

Probability generating function of $N_k(G)$ is

$$\begin{aligned} E(z^{N_k(G)}) &= \prod_{\text{all } k\text{-cubes}} \{ \mathbb{P}[E(C)] + (1 - \mathbb{P}[E(C)]) z \} \\ &= \prod \{ e^{-\mu(C)} + (1 - e^{-\mu(C)}) z \} \quad \text{in } |z| < 1. \end{aligned}$$

For $0 \leq z < 1$ by monotone convergence thm.

$$E(z^{N(G)}) = \lim_{k \rightarrow \infty} \prod \{ z + (1 - z) e^{-\mu(C)} \}$$

By this inequality $z + (1 - z) e^{-\mu} \geq e^{-(1-z)\mu}$.

$$\therefore E(z^{N(G)}) \geq \lim_{k \rightarrow \infty} \prod e^{-(1-z)\mu(C)} \geq e^{-(1-z)\mu(G)}$$

Let $z \rightarrow 1 \Rightarrow N(G)$ is finite a.s, ($\because G$ bounded $\Rightarrow \mu(G) < \infty$)

Fact) k -cubes in a bounded region, $\mu(C) \rightarrow 0$ uniformly as $k \rightarrow \infty$

(pf)

If it is false, $\exists \delta > 0, \forall k$ k -cubes $C \subset C_0$ $\mu(C) > \delta$

Call a k -cube 'bad' if $\forall l > k$ l -cube $C' \subset C$ & $\mu(C') > \delta$

By assumption C_0 is bad

Divide C_0 into 2^d 1-cubes $\rightarrow$ at least one is bad

...

$C_0 \supset C_1 \supset \dots$ where C_k is bad k -cube

$\Rightarrow \forall k, \mu(C_k) > \delta$

But C_k is k -cube with side 2^{-k}

$\cap C_k$: empty or single point & $\mu(\cap C_k) > \delta \Rightarrow$ contradiction by non-atomic $\square$

$\therefore$ For sufficiently large k , $\mu(C) \leq \delta$ for all k -cubes in G

If $\mu \leq \delta$ and z is fixed

$$e^{-(1-z)\mu} \leq z + (1-z)e^{-\mu} \leq e^{-(1-z)\psi(\delta)\mu} \quad (\psi(\delta) = \frac{z}{\delta + z} + z \leq \frac{\mu^2 z \delta \epsilon}{\epsilon^2 (e^{\mu-1})})$$

$$\therefore e^{-(1-z)\mu(G)} \leq E(z^{N(G)}) \leq e^{-(1-z)\psi(\delta)\mu(G)}$$

Let $\delta \rightarrow 0$ ($\psi(\delta) \rightarrow 1$)

$$E(z^{N(G)}) = e^{\mu(G)(z-1)} \quad \text{for all } 0 \leq z < 1$$

$\therefore N(G)$ has a Poisson distribution with mean $\mu(G)$

Let $G_1, \dots, G_m$: disjoint bounded open sets.

No k -cube can be contained in more than one G_j

$\Rightarrow N_k(G_1), \dots, N_k(G_m)$: independent

$\therefore N(G_1), \dots, N(G_m)$: independent ($k \rightarrow \infty$)

Also you can extend to general hael sets.

